

A decorative graphic on the left side of the slide, consisting of white lines and circles on a blue gradient background, resembling a circuit board or a stylized tree structure.

ELECTRONIC CIRCUITS

ICT 1202

References

1. A first course in Electronics (1st edition), Anwar A Khan
2. The Art of Electronics (2nd Edition), Paul Horowitz , Windfield Hill
3. Electronics and Electronic Systems (1st Edition), George H Olson

Course Capsule

- Introduction to electronics and electronic systems, introduction to semiconductor and devices like diodes, BJT, FET, MOSFET, Rectifier and Filters, introduction to transistor biasing. Small signal transistor amplifiers.
- Digital concepts: basic logic operations, logic functions, introduction to ICs. Logic Gates and Circuits: AND, OR, XOR. Boolean Algebra and Logic Simplification: Boolean Operations, De Morgan's theorem, Karnaugh Map. Combinational logic: NAND and NOR gates, adders, comparators, decoders, encoders, multiplexers, de-multiplexers. Sequential logic: flip-flops, timers, counters, shift registers, memories.

Intended Learning Outcomes

1. Explain the difference in conductivity of solids using energy band diagrams and derive equations for conductivity of semiconductors.
2. Explain the difference between intrinsic and extrinsic semiconductors and fabricating **p- type** and **n- type** semiconductor using intrinsic semiconductors
3. Explain the formation, characteristics and charge flow of a **pn** junction.
4. Derive equations and explain the variation of charge, electric field and potential across a **pn** junction
5. Discuss the applications of **pn** junctions.
6. Analyze circuits using Norton and Thevenin's theorems.
7. Explain the basic characteristics, biasing and applications of BJ transistors.
8. Explain the properties and uses of OP amps.

Basic concepts

- Electric current
- Voltage
- Voltage source
- Current source
- Ohms law
- Resistance, Reactance and Impedance
- Resistor
 - Parallel connection
 - Serial connection
- Kirchhoff's law
- Thevenin's theorem
- Norton's theorem

- Electric current

- An electric current is created by the flow of electronic charges. In solids, charges is often carried by electrons. In liquid and gases, charge is often carried by ions. In general, any particle with an electric charge and is free to move, is called “*Charge carries*”.
- Units – Coulombs (C).
- Electron has charge – 1.602×10^{-19} C
- Flow of one coulomb charge, across a surface in a second is equal to one Ampere (A).

- Voltage

- It is the amount of electric potential difference between two points. This difference can be caused by static electric field, electrochemical reaction, electromagnetic induction or any other.
- Moving 1 Coulomb of charge across two points with 1 Volts of electric potential difference, requires 1 Joule of work. The amount of work is independent of path.

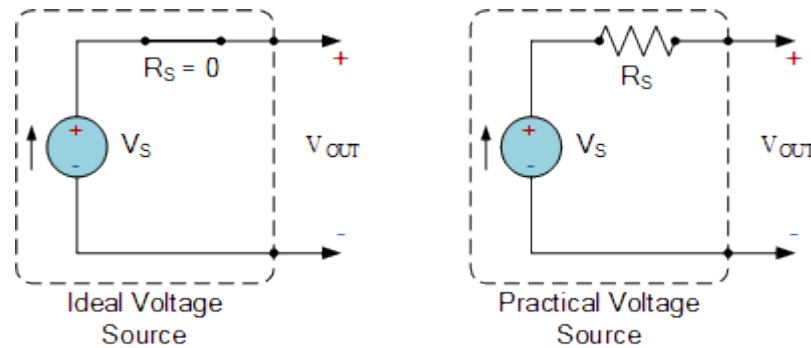
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- Voltage source

- An ideal voltage source can maintain a fixed voltage across its two terminals, regardless of the current drawn from it (i.e. independent of the load resistor).

The internal resistance of an ideal voltage source is zero.

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- However, a real world voltage sources have maximum current limit. If more current is drawn from the source either the voltage will drop or the source will be damage.



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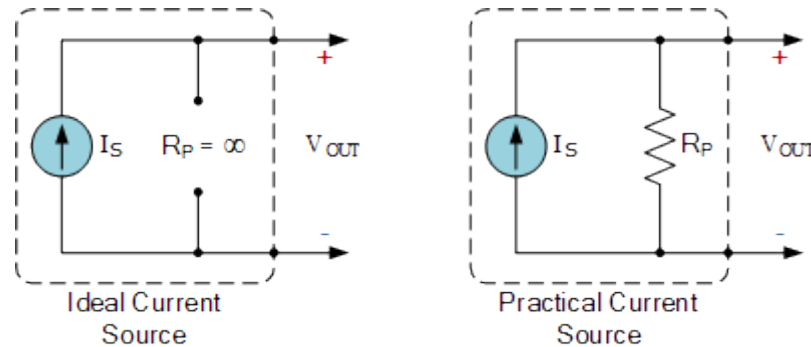


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Voltage is not everything !!

- Current source

- An ideal current source will deliver (or absorb) constant current across a load resistor regardless of load resistance. If the voltage resistance increase, applied voltage will automatically increase to maintain the same amount of current. The internal resistance of an ideal current source is infinite.



- Ohms law

- Ohm's law states that the current through a conductor between two points is directly proportional to the voltage across the two points.

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$$V = IR \text{ where, } R \text{ is resistance measured in ohms } (\Omega)$$

- Although Ohm's law is the most famous and most widely used law in electronic, it cannot be apply to nonlinear elements such as Diodes, incandescent bulbs.
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- Resistance, Reactance and Impedance

- Some components have 'resistance' that change with the frequency of the applied voltage. This is called 'reactance'.

- For example

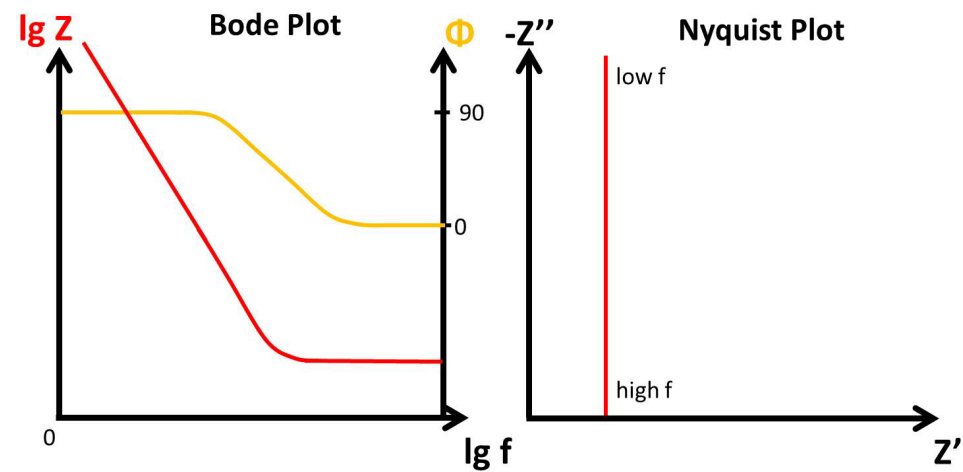
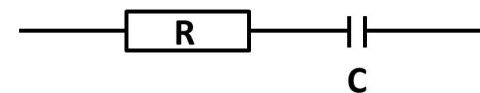
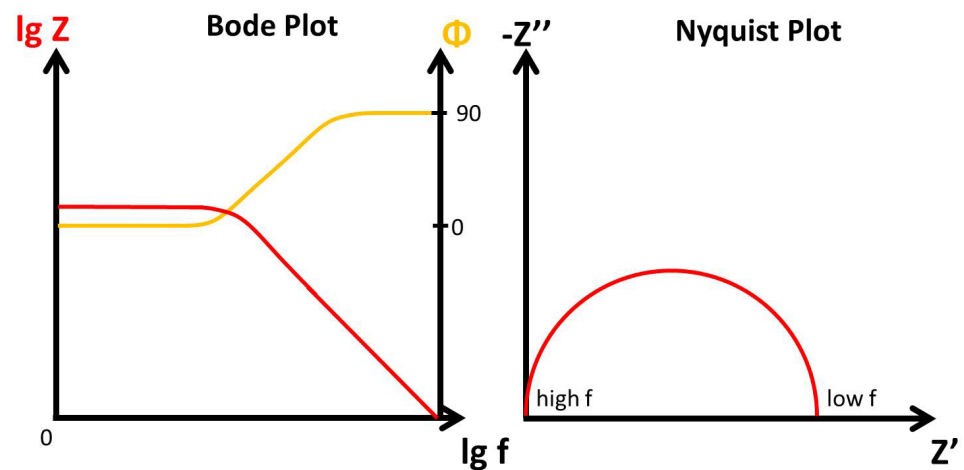
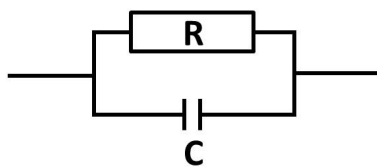
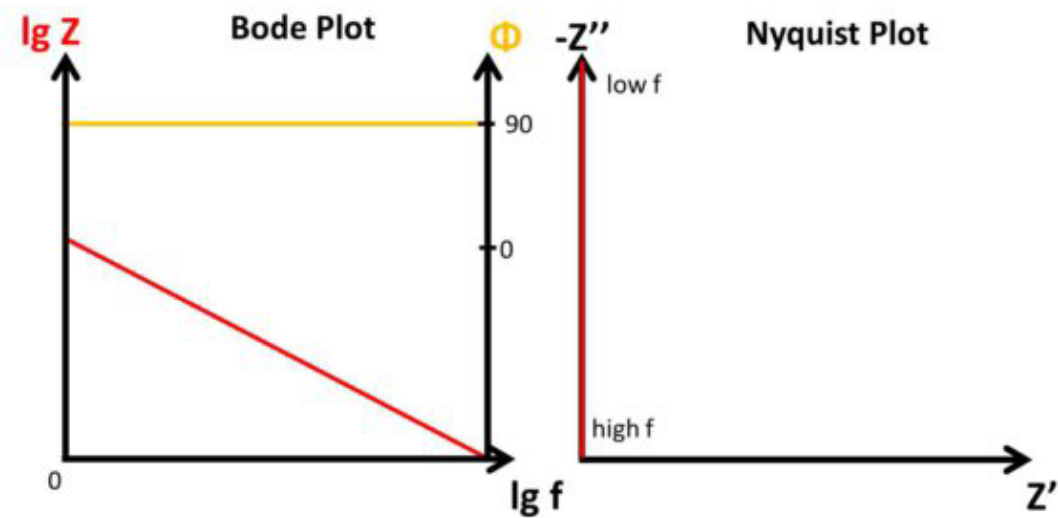
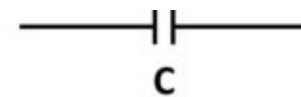
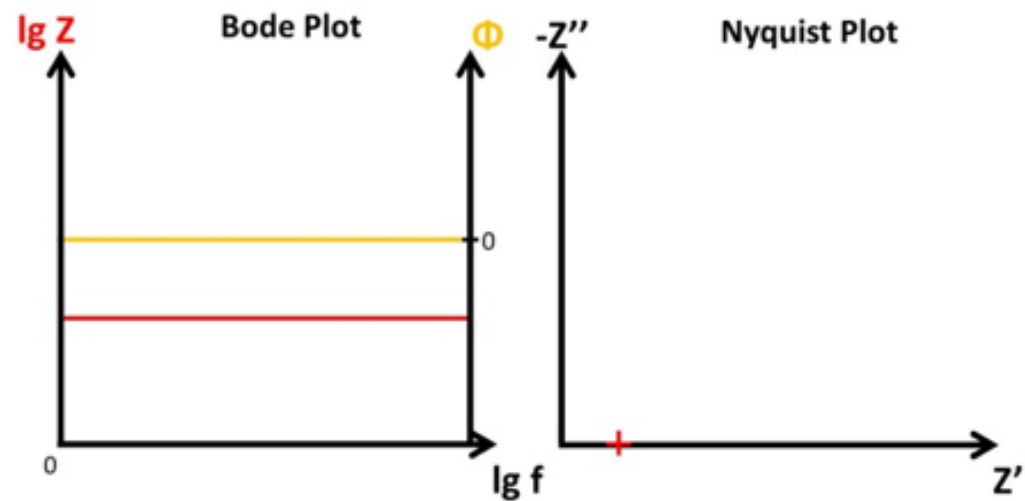
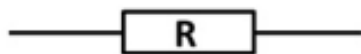
- Resistance of capacitor $X_c = -\frac{1}{\omega C}$
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- Resistance of inductor $X_L = \omega L$
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- Where ω , ($\omega = 2\pi f$) is the frequency of applied signal.
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- Impedance (Z) is the total opposition provided by circuit to a given signal. It may contain both resistant and reactance. However reactance component need to be multiplied by 'i' ($i^2 = -1$) and cannot be added with resistance.

$$Z = R + iX$$

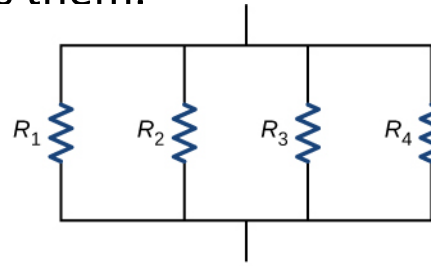


- Resistors

- Resistors are used to limit the current. There are different types of resistors.

- Parallel connection සමාන්තර සම්බන්ධය

- In parallel connections, components have shared connections. Therefore, all components have the same voltage across them.



- All the resistors (from R1 to R4) receive the same voltage. Net current of the circuit is the summation of current in each branch.

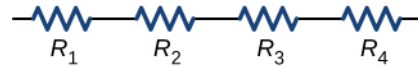
$$V_{R1} = V_{R2} = V_{R3} = V_{R4}$$

$$I = I_{R1} + I_{R2} + I_{R3} + I_{R4}$$

$$\frac{1}{R} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3} + \frac{1}{R4}$$

- Serial connection ශ්‍රේණිගත සම්බන්ධතාව

- In series connection, each component is connect to the adjacent component and only the terminal at the end are available.
- Equal current flows across all the resistors.
- Net voltage across two terminal is equal to the summation of voltage across each component.



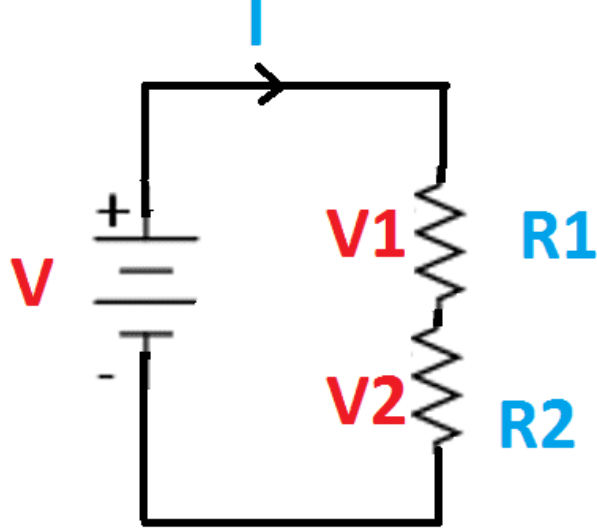
$$I_{R1} = I_{R2} = I_{R3} = I_{R4}$$

$$V = V_{R1} + V_{R2} + V_{R3} + V_{R4}$$

$$R = R_1 + R_2 + R_3 + R_4$$

- This gives rise to the concept of voltage divider. The voltage division rule states that voltage drop across any resistance (or combination of resistors) in a series current is equal to the ratio of that resistance to the effective resistance multiplied by the source voltage.

Voltage Divider



$$I = \frac{V}{R} = \frac{V}{R_1 + R_2}$$

$$V_2 = IR_2$$

$$V_2 = \left(\frac{R_2}{R_1 + R_2} \right) V$$

- **Kirchhoff's law**

- When resistors connected in serial, parallel or as a combination of them, it is easy to calculate the net resistance and the current passing through each branch. However, there are circuits that cannot be simplified in that manner. We can use the Kirchhoff's law to solve such problems.

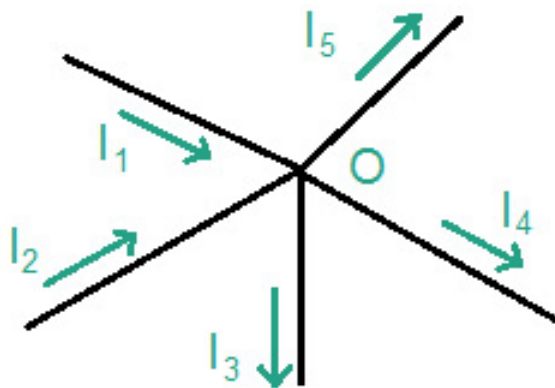
- First law – Kirchhoff's current law

- At any instant, the algebraic sum of the current at a junction in a network is zero.

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- Second law – Kirchhoff's voltage law

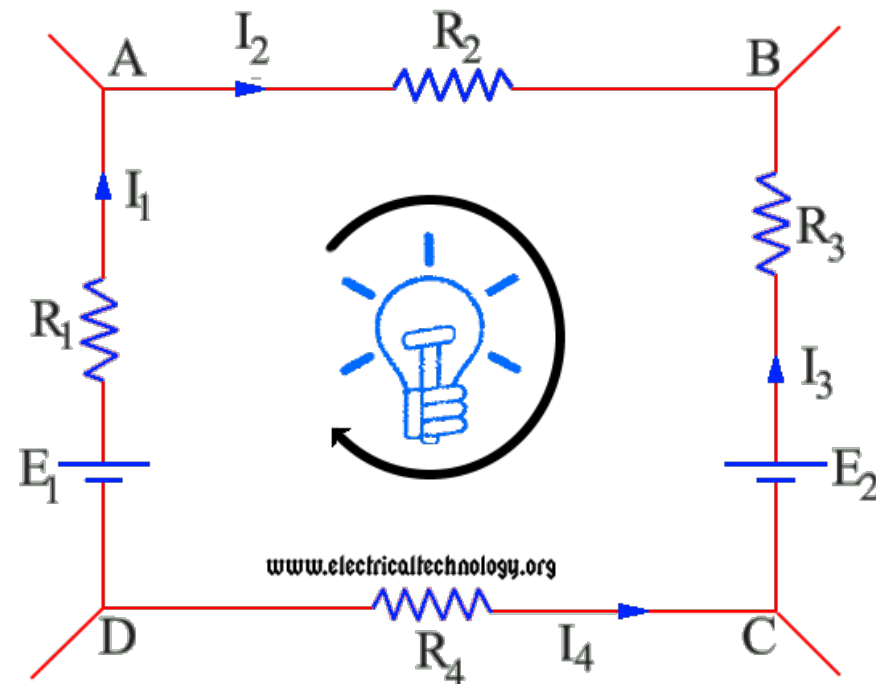
- In a closed loop, the algebraic sum of all the e.m.f.'s acting round the loop is equal to the algebraic sum of the voltage drops round the loop.



Applying KCL:-

$$I_1 + I_2 - I_3 - I_4 - I_5 = 0$$

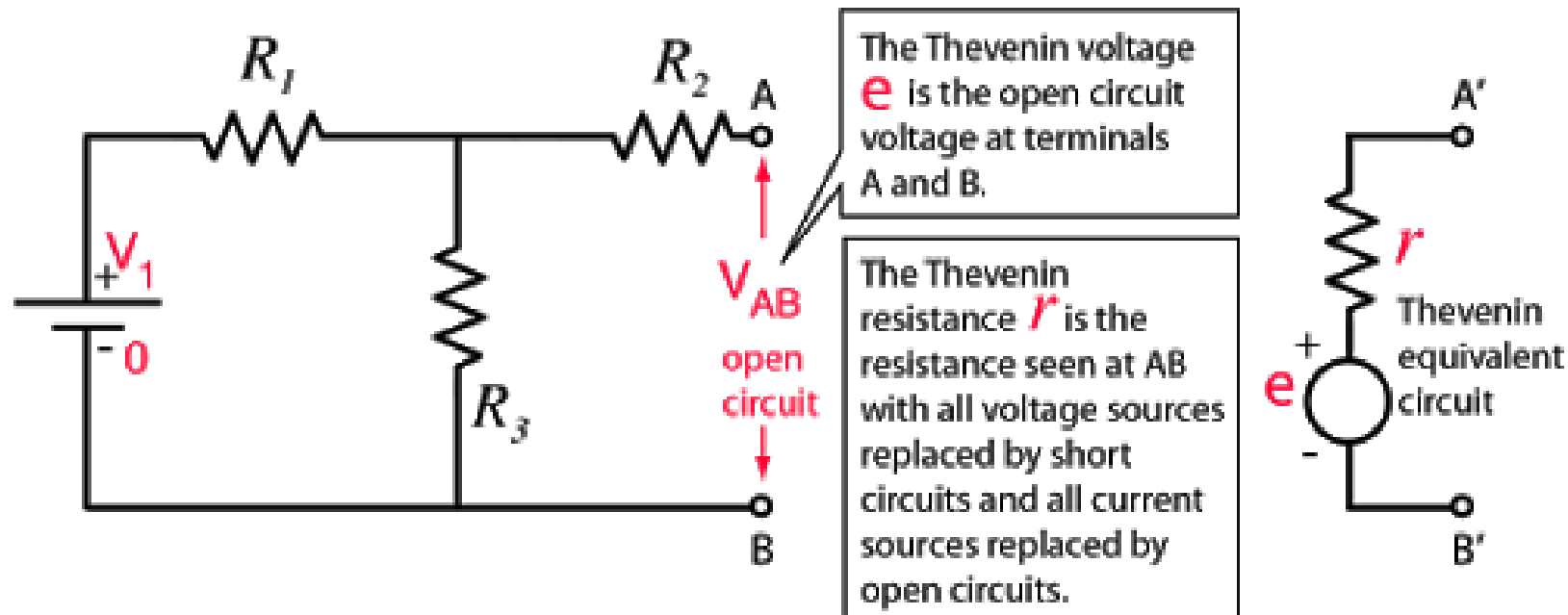
$$I_1 + I_2 = I_3 + I_4 + I_5$$



$$E_1 - E_2 = I_1 R_1 + I_2 R_2 - I_3 R_3 - I_4 R_4$$

- Thevenin's theorem

- The current which flows in any branch of a circuit is the same as that which would flow in the branch if it were connected across a source of electrical energy, the e.m.f. of which is equal to the potential difference which would appear across the branch if it were open-circuited, and the internal resistance of which is equal to the resistor which appears across the open circuited branch terminals.



- Norton's theorem

- The Norton's theorem is actually a restatement of Thevenin's theorem using equivalent current sources instead of the equivalent voltage source.
- The current which flows in any branch of a network is the same as that which would flow in the branch if it were connected across a source of electric energy that short-circuit current of which is equal to the current that would flow in a short-circuit across the branch, and the internal resistance of which is equal to the resistance which appears across the open circuited branch terminal.

